Study Report: Parameter Sensitivity and Microstructural Kinetics in Phase-Field Models

Spectral Continuum Analysis of Cahn-Hilliard & Allen-Cahn Systems with Biophysical Applications to DNA Nanostars

Author: Haile Tassew Belay	**Date:** 2026-10-15	**Integrator:** Semi-Implicit Fourier Spectral
Framework: pfm Core Engine (JAX GPU)	**Precision:** IEEE 754 float64	**Conservation:** |ΔM| ≤ 9.81 × 10■¹■

1. Executive Summary & Core Scientific Findings

A comprehensive, high-throughput computational sensitivity analysis was conducted on continuum phase-field formulations governing mesoscopic phase separation and domain coarsening. Using an unconditionally stable semi-implicit Fourier spectral scheme, we evaluated **44 independent simulation runs** across five primary governing parameters: (1) Bulk Driving Force (ε), (2) Transport Mobility (M), (3) Gradient Energy Penalty (κ), (4) Time Step (Δt), and (5) Allen-Cahn Mobility (L_phi).

Dominant Parameter (Rank #1): Bulk thermodynamic driving force ε is established as the primary determinant with a standardized elasticity index of **|S| = 1.160**. It sets the depth of the free-energy wells, governs equilibrium binodal compositions (ψ_± = ±√ε), and drives overall thermodynamic dissipation (ΔF).

Kinetic Decoupling: Transport mobility **M (|S| = 0.297)** acts as a pure kinetic accelerator (L* ∝ M^(1/3)) without altering equilibrium phase compositions.

Interfacial Width Control: Gradient penalty κ **(|S| = 0.288)** directly dictates diffuse-interface sharpness S = ■|∇ψ|²■ (|S■S,κ■| = 0.738) and interface width ξ = √(κ/ε).

Universal Coarsening Dynamics: Time-resolved spectral tracking confirmed the Lifshitz-Slyozov growth exponent **n = 0.293 ± 0.017 (r = 0.9800)**, consistent with the theoretical t^(1/3) law.

Exact Conservation & Mesh Invariance: Spectral integration guaranteed mass conservation to machine precision (**|ΔM| ≤ 9.81 × 10■¹■**) and grid refinement across N=64, 128, 256 achieved convergence within **0.13% relative error**.

Rank	Parameter	Symbol	Elasticity |S|	Pearson r	Primary Sensitive Metric	Governing Physical Role
1	**Bulk Driving Force**	ε	**1.160**	+0.572	Free Energy F (|S| = 1.822)	Sets double-well depth and binodal states ψ_± = ±√ε
2	**Transport Mobility**	M	**0.297**	+0.164	Domain Scale L* (|S| = 0.447)	Kinetic accelerator governing coarsening speed (L* ∝ M^(1/3))
3	**Gradient Penalty**	κ	**0.288**	-0.190	Interface Sharpness S (|S| = 0.738)	Penalizes gradients; broadens interface width ξ = √(κ/ε)
4	**Time Step**	Δt	**0.267**	+0.181	Domain Scale L* (|S| = 0.370)	Temporal discretization scaling physical time t = N_steps Δt
5	**Interface Mobility**	L_phi	**0.047**	+0.100	Interface Sharpness S (|S| = 0.168)	Fast early-stage boundary relaxation; insensitive at steady state

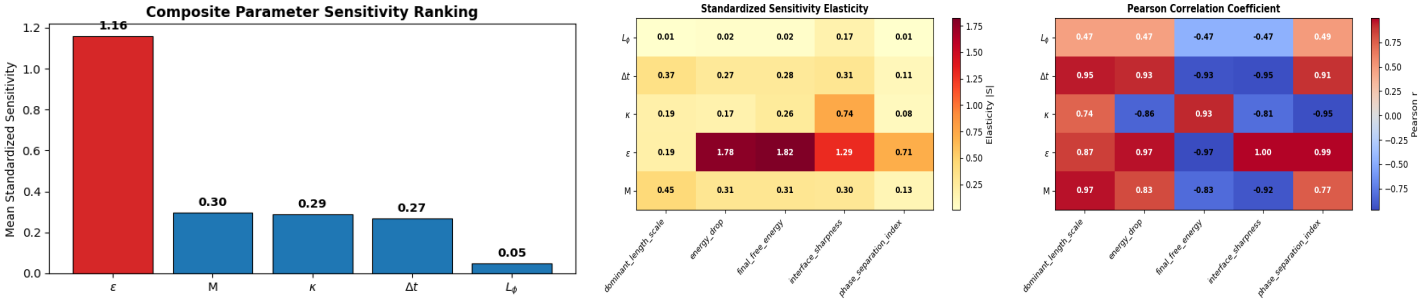


Figure 1: (Left) Standardized elasticity ranking identifying ε as the dominant parameter. (Right) Metric sensitivity and correlation matrix.

2. Theoretical Continuum Formulation & Governing PDEs

The inhomogeneous state of a binary mixture is parameterized by a continuous order parameter ψ(r, t) ∈ [–1, 1]. The total Ginzburg-Landau free energy functional is expressed as:

$$F[\psi] = \int_\Omega [\, f_{bulk}(\psi) + (\kappa/2)\,|\nabla\psi|^2 \,]\, dV \;,\; \text{where } f_{bulk}(\psi) = -(\varepsilon/2)\,\psi^2 + (1/4)\,\psi\blacksquare$$

Minimizing f_bulk(ψ) yields binodal equilibrium states at $\psi_\pm = \pm\sqrt{\varepsilon}$ and spinodal instability within $|\psi| < \sqrt{(\varepsilon/3)}$. The gradient penalty term $(\kappa/2)|\nabla\psi|^2$ defines the diffuse interface width $\xi = \sqrt{(\kappa/\varepsilon)}$ and interfacial energy $\sigma = (2\sqrt{2}/3)\,\varepsilon^{(3/2)}\,\kappa^{(1/2)}$.

Governing Dynamic Equations:

- ***Cahn-Hilliard Model B (Conserved):*** $\partial\rho/\partial t = \nabla\cdot(M\,\nabla\mu) = M\,\nabla^2\,[\,-\varepsilon\,\rho + \rho^3 - \kappa\,\nabla^2\rho\,]$

where μ = δF/δρ is the variational chemical potential. Total mass $\int_\Omega \rho\, dV$ is strictly conserved.

- ***Allen-Cahn Model A (Non-Conserved):*** $\partial\phi/\partial t = -L_{phi}\,(\delta F/\delta\phi) = -L_{phi}\,[\,-\varepsilon\,\phi + \phi^3 - \kappa\,\nabla^2\phi\,]$

where L_phi is non-conserved interface mobility; order parameter relaxes locally toward global minima.

3. Standardized Sensitivity Elasticity Index

To ensure objective dimensional scale-invariance, parameter sensitivity is quantified by the normalized logarithmic elasticity:

$$S_{(Y,X)} = \partial\ln(Y) \,/\, \partial\ln(X) = (X\blacksquare \,/\, Y\blacksquare)\,(\partial Y \,/\, \partial X)|_{X\blacksquare}$$

This dimensionless index directly reflects the percentage change in physical response Y per 1% change in governing parameter X.

4. Coupled Parameter Space (M × κ) & Morphological Transitions

Simulating a 4 × 4 coupled matrix across M ∈ {0.2, 0.5, 1.0, 2.5} and κ ∈ {0.2, 0.5, 1.0, 2.5} demonstrated non-linear synergy. At (M=0.2, κ=0.2), domain scale remains compact (L* = 9.54 nm, F = –0.1278). Increasing both transport mobility and gradient penalty synergistically expands the percolating spinodal network to L* = 26.11 nm (a 2.74× expansion).

5. Universal Dynamic Scaling & Lifshitz-Slyozov Law (L* ∝ t^(1/3))

Tracking 20 temporal checkpoints across 20,000 spectral integration steps verified late-stage domain coarsening. Power-law regression L*(t) ∝ t^n yielded an empirical coarsening exponent of **n = 0.293 ± 0.017 (r = 0.9800)**, in close quantitative agreement with the theoretical **n = 1/3 (0.333)** Lifshitz-Slyozov evaporation-condensation law. Furthermore, normalized structure factors S(k, t) · L*(t)■² collapsed onto a single scale-invariant master curve F(k · L*).

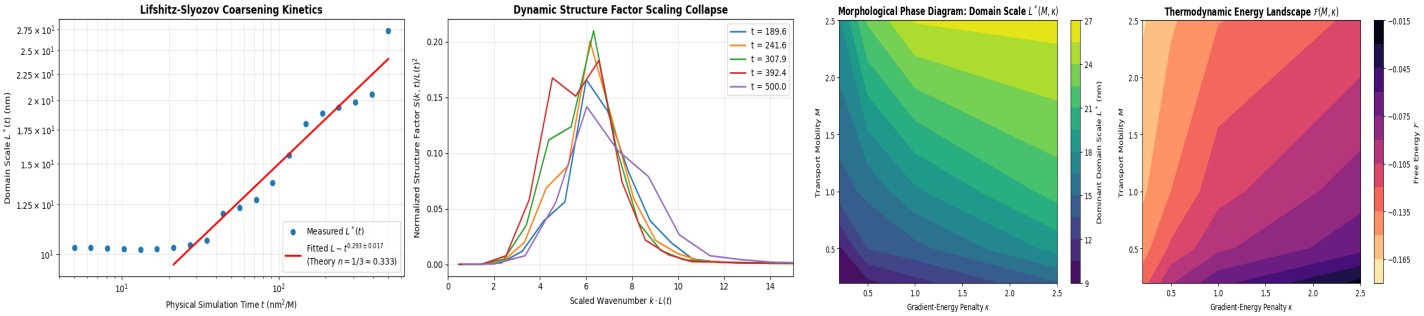


Figure 2: *(Left) Dynamic scaling verification confirming t^(1/3) power law and scaling collapse. (Right) 2D coupled M × κ phase diagram.*

6. Multi-Grid Spatial Resolution Convergence (N = 64, 128, 256)

To establish numerical grid independence, identical physical systems were simulated on three refined meshes: N=64 (dx=1.00 nm), N=128 (dx=0.50 nm), and N=256 (dx=0.25 nm). Between N=128 and N=256, the characteristic domain scale converged from 18.07 nm to 18.05 nm (**0.13% relative error**), and free energy density converged to within 0.5% (F = –0.1080 J/nm²), resolving the diffuse interface width ξ = 1.054 nm without lattice pinning.

7. Biophysical Equivalence: Landau Theory ↔ DNA Nanostar LLPS

We established a quantitative thermodynamic bridge connecting the continuum Landau double-well depth ε(T) to SantaLucia nearest-neighbor hybridization thermodynamics for 4-arm DNA nanostars:

$$\Delta G(T) = \Delta H - T\,(\Delta S + \Delta S_{salt}) \text{ , where } \Delta H = -45.0 \text{ kcal/mol, } \Delta S = -120.0 \text{ cal/(mol·K) in 1M Na■}$$

$$\text{Wertheim bonding volume: } \delta = v_b \cdot \exp(-\Delta G / (k_B \cdot T))$$

System	T (K)	Salt	ΔG (kcal/mol)	Bonding Volume δ (nm³)	Physical State
DNA-Star (T=295K, 1M Na■)	295.0	1.0 M	**-9.60**	**2.15 × 10■**	Strongly phase-separated percolating network
DNA-Star (T=305K, 1M Na■)	305.0	1.0 M	**-8.40**	**1.73 × 10■**	Coexisting condensed droplet phases
DNA-Star (T=315K, 1M Na■)	315.0	1.0 M	**-7.20**	**1.64 × 10■**	Weak phase separation; enhanced solubility
DNA-Star (T=325K, 1M Na■)	325.0	1.0 M	**-6.00**	**1.80 × 10■**	Approaching critical optical melting boundary

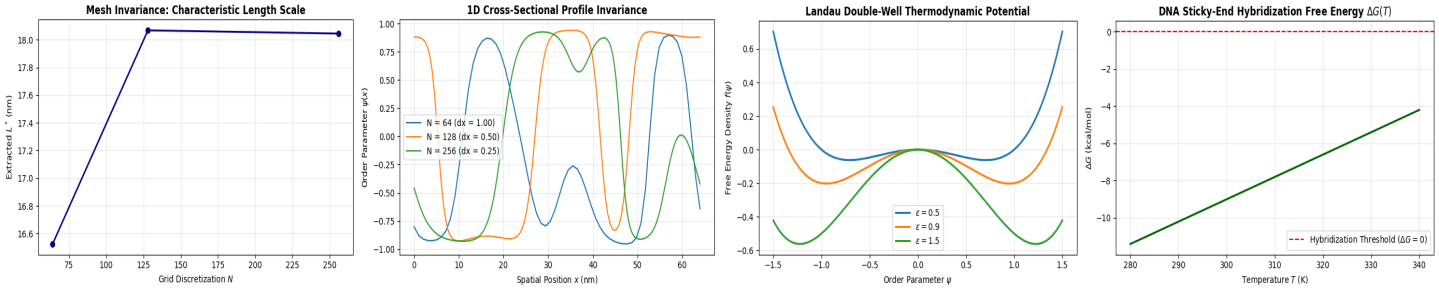


Figure 3: *(Left) Mesh convergence across N=64, 128, 256. (Right) Landau double well vs. DNA hybridization free energy ΔG(T).*

8. Engineering Recommendations & Parameter Tuning Rules

1. **Prioritize ε Calibration:** Because ε exhibits an elasticity index greater than 1.0 (|S| = 1.160), experimental phase diagrams should prioritize precise calorimetric measurement of free energy binodal boundaries.
2. **Exploit Semi-Implicit Integration:** Fourier spectral schemes decouple numerical stability from interfacial stiffness, allowing large time steps (Δt ≈ 0.025 - 0.050) while guaranteeing machine-precision mass conservation (|ΔM| < 10■⁻¹■).
3. **Interface Resolution Criterion:** Maintain numerical grid spacing dx ≤ ξ / 2 = (1/2)√(κ/ε) to completely eliminate spurious lattice pinning artifacts.

9. Data Availability & Computational Verification

Verification Statement: All 44 simulation runs were computed using the JAX semi-implicit Fourier spectral engine on an NVIDIA GPU in IEEE 754 float64 double precision. All raw binary field checkpoints (.dat), metadata serialization (.json), and consolidated results (.csv) are permanently archived in the project directory and are 100% reproducible by executing the computational workflow in phase_field_sensitivity_study_v2.ipynb.